# Pulse CBD — Project Report and Handover

**As of:** Thursday 5 November 2026, 4:30 pm (Sydney), during the last recorded PM peak before the showcase
**Showcase:** Friday 6 November 2026
**Feature freeze:** 30 October 2026. Changes after it, all at the owner's request: background map (4 Nov), street-colour fixes and direction modes (4 Nov), replay day picker and full colour definitions on the tablet (5 Nov)
**Build:** branch `w12merge` (the owner's `95c722a` merged with the teammate's corner fit, 5 Nov), 208 automated tests passing
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## How to read this report

This report is written for someone who has never seen the project: a teammate picking it up, an assessor, or a new Claude Code session with no memory of earlier work. It assumes no knowledge of the code, of Transport for NSW's data, or of the decisions made along the way. Every technical term is explained the first time it matters, and §3 is a glossary to come back to.

It is organised in five parts:

- **Part I — Understanding the project** (§1–3): what Pulse CBD is, what someone standing at the model sees, the course and physical setup it was built for, and the ideas needed to follow the rest.
- **Part II — How it works** (§4–8): the parts of the system and how data moves through them, every data source, every feature, exactly how every number is calculated, and what every colour means.
- **Part III — Why it is built this way** (§9–11): the main design decisions and the alternatives rejected, the analysis comparing our own congestion measure with Transport for NSW's, and how the system was tested.
- **Part IV — Running it** (§12–14): how to start, stop, record, calibrate and troubleshoot it, the code repository, and the exact state of the laptop today.
- **Part V — Assessment** (§15–20): limitations to state honestly in the final report, remaining work, suggestions, open decisions, the timeline, and a map of the key files.

The appendices list every display setting, every server address and every configuration value. The full reasoning behind each decision, in the order decisions were made, is in `docs/DECISIONS.md`; this report summarises it.

**For a new Claude Code session, the essentials:**

| Thing | Where |
|---|---|
| Project folder | `~/pulse-cbd` (Python 3.13 virtual environment in `.venv`; activate with `source .venv/bin/activate`) |
| Branch the laptop runs | `w12merge`, pushed to GitHub (latest commit: `git log -1`) |
| GitHub remote | `group-repo` → `https://github.com/0rangeGoose/Civil3704-group-6` (private) |
| Decision log | `docs/DECISIONS.md` (newest first) |
| How to run everything | `README.md` |
| Original project brief | `~/Downloads/pulse-cbd-claude-code-prompt.md` |

The owner's working rules (from `~/.claude/CLAUDE.md` and instructions given in sessions):

- Ask before any git commit unless told to commit. New work goes to `w12merge`. **Never push to `integration`** without an explicit instruction.
- Check the changes for the TfNSW API key (kept in `.env`) before every push.
- Use the `gh` command-line tool for GitHub.
- When instructions are unclear or conflicting, stop and ask; never override or skip an instruction without asking.
- Explain in plain language; the owner asks "what do you mean by…" when jargon slips in.
- After changing a page, check it in Chrome; after changing the server, restart it (`launchctl kickstart -k gui/$(id -u)/com.pulsecbd.server`). Open pages need a reload to pick up new code.

```
cd ~/pulse-cbd && git status -sb && git log --oneline -3
lsof -nP -iTCP:8765 -sTCP:LISTEN           # is the server running?
curl -s http://127.0.0.1:8765/health        # vehicles, feed age, API quota use, playback mode
tail -5 data/recordings/recorder.log        # scheduled peak recordings
source .venv/bin/activate && pytest -q      # 208 tests should pass
```

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# Part I — Understanding the project

## 1. What Pulse CBD is

### 1.1 In one paragraph

Pulse CBD turns a physical scale model of inner Sydney into a live map of its public transport. A projector mounted above the model shines an image straight down onto it. That image shows every train, metro, light rail vehicle, ferry and bus currently inside the model's area — typically 270 to 490 of them at once (487 at 4:27 pm on 5 November) — as a small moving arrow on the street or track it is really on, updated every 15 seconds from Transport for NSW's live data. On top of the vehicles, coloured layers answer planning questions: which streets are congested right now, where buses are full, where buses typically lose time, where lots of jobs are served by little public transport, and how one peak hour compares with another. It can show what is happening now, or replay any recorded morning or evening peak as a sped-up time-lapse. Visitors control it from a tablet beside the model; the projected image itself contains only map content.

### 1.2 What a visitor experiences

Someone walking up to the model during the showcase would see, in a typical demonstration:

1. **The network moving.** Hundreds of arrows travel along the model's streets and rail lines. Bigger arrows are trains, metro and light rail; smaller ones are buses and ferries. Each leaves a short fading tail so its movement is visible even between updates. By default each arrow is coloured by punctuality: green on time, amber running late, red very late or cancelled, grey when there is no schedule information to compare with.
2. **A peak hour in five minutes.** The team taps *Replay mode* on the tablet, picks a day and *AM peak*. The model jumps to 7:00 that morning and plays it back 25 times faster than real time, so two hours of the morning rush pass in under five minutes. The presenter can pause, jump forwards or back, and point out where buses bunch up.
3. **Coloured streets.** Turning on *Traffic* colours the streets buses are using: green where they are running smoothly, yellow stop-and-go, orange congested, red severely congested. Turning on *Crowding* colours

them by how full vehicles are. Other street layers show typical bus loads and where buses typically fall behind the timetable, from Transport for NSW's bus ticketing data.

4. **Coloured areas.** Turning on an area layer shades the whole model in patches — either Transport for NSW's "travel zones" or a regular 150 m grid — by how many jobs are there, by how well-served those jobs are by public transport (the *service gap*), by how many Opal card taps were made there, or by a combined *pressure* score.
5. **Stops with a history.** *Stop markers* show every bus, train, light rail and ferry stop on the model. Turning on *Stop history* colours each stop by how late vehicles typically are when they reach it (median or mean), or, with the crowding colours on, by how often they arrive with few seats or fuller. It draws on every stop visit recorded live since 30 October — 865,907 by the afternoon of 5 November — and a stop needs at least 20 visits before it is coloured. The teammate's notes (as of 4 November) report that at 610 of 662 stops (92%) crowded vehicles arrived later than seated ones: an association, not proof that crowding causes delay. A stop's colour shows lateness *inherited* from earlier in the trip, not only delay caused at that stop (§7.13).
6. **One route at a time.** Typing route names into the *route filter* — for example 333, T1 or L2, up to eight at once — makes those routes' vehicles, line and stops stand out, each in its own fixed colour, while everything else fades back. It is the easiest way to follow one bus route through the peak.
7. **Both directions of a street.** By default each street shows one colour for both directions. *Street directions* switches to *Side by side*, with each direction drawn on its own side of the street, or to *To city* or *From city*, showing only traffic heading towards or away from Town Hall. On the 4 November AM peak replay, for example, it showed a congested inbound side right next to a clear outbound one, a difference the one-colour view averages away.
8. **A map underneath, on a screen.** The model already has streets, so the background is plain by default. For a laptop or flat-screen demonstration, *Background map* adds faint roads, rail, water and parks (*Detailed*), or those plus suburb, station, landmark and street names (*Full*).
9. **A legend that explains everything.** The tablet's *What the colours mean* section explains every colour currently shown, with the exact numbers behind it — for example, "On time: up to 5 min behind the timetable for trains, buses and ferries; 2 min for metro and light rail".

### 1.3 What it is for

The course asks for a transport planning tool, and the model is the canvas. Pulse CBD's purpose is to make the network's performance *visible in place*: congestion, crowding and access shown on the actual streets and districts of a model people can walk around, rather than in charts. It answers questions such as:

- Where are buses delayed during the morning peak, and does it match where Transport for NSW says traffic is congested?
- Which streets carry the fullest buses?
- Which parts of the inner city have many jobs but few nearby public transport departures?
- How does today's peak compare with last Tuesday's?

### 1.4 Status at a glance

The project was planned on 25 September as seven steps. Six are finished; the seventh needs the physical model.

| Step (agreed 25 Sep) | What it means | Status |
|---|---|---|
| 1. Footprint, projector view, tablet controller, scheduled recording | The basic system: the right map area, the projected image, a tablet to | Done |

| Step (agreed 25 Sep) | What it means | Status |
|---|---|---|
| | control it, automatic recording of peaks | |
| 2. Traffic and crowding on streets | Colour streets by Transport for NSW's live traffic and crowding fields | Done; improved 4 Nov (§7.5) |
| 3. Replay and time-lapse | Play back recorded peaks, sped up | Done; pause and scrub added 30 Oct, any-day picker 5 Nov |
| 4. Jobs, service gap, Opal 2016 | Area layers from employment, timetable and ticketing data | Done |
| 5. Combined "pressure" score | One area score combining several inputs, with adjustable weights | Done |
| 6. Our own congestion measure, compared with Transport for NSW's | An independent measure we can explain, and a statistical comparison | Done; the method changed and agreement turned out weak (§10) |
| 7. Testing on the model, report figures, rehearsal | Calibrate onto the real model and rehearse the showcase | **Partly.** The teammate's 5 Nov README says the system is running on the physical model, aligned with their new corner fit (§12.4). Report figures and a rehearsal aren't recorded (§16) |

Beyond the plan, the project also has: bus load and time-lost layers from Transport for NSW's BOAM data; polling that automatically stays within the daily data quota; a refreshed timetable; a teammate's stop history, route filter and direction arrows (merged 30 Oct); an optional background street map; and full colour definitions on the tablet.

- **Tests:** 208 automated tests run in about 2 seconds without internet access.
- **Data recorded:** 16 weekday peaks (15 of them at least 98% complete), plus weekends, nights and middays — 26 replayable sessions in total. The Thursday 5 November PM peak is recording as this is written.
- **Biggest risks:** every layer hasn't been checked on the model in this record (alignment has, per the teammate); the showcase network setup is untested (§16); a Wi-Fi drop during a live demo would freeze the vehicles; every recording is on one laptop disk.

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## 2. Background

### 2.1 The course and the team

This is the CIVIL3704 group project for Group 6, with a final showcase on **Friday 6 November 2026**, where the team presents to assessors and visitors then try the installation themselves. Team roles were never formally divided.

Two pieces of software came out of the team:

- **Pulse CBD** (this codebase), built by the owner, Yoseph Kim, starting as a prototype in August.
- **A separate web dashboard** built by teammates (branches `mid-project-demo`, `test-pipline`), which computed its own congestion and demand scores on a map.

On 25 September the team agreed to make Pulse CBD the single codebase, keep the dashboard's *direction* (a planning tool about congestion, demand and peak comparison), and rebuild its calculations, which had flaws

(§9.1). Teammate **0rangeGoose** later added stop history, fixes to how delays are read, a route filter and direction arrows to Pulse CBD itself (29–30 Oct); that work is merged in (§13).

### 2.2 The physical setup

- **The model.** A tabletop scale model of Sydney's inner city. The team surveyed its corners: north-east –33.835123, 151.275993 and south-west –33.894723, 151.132966. That is about **13.2 km east–west by 6.6 km north–south**, roughly two to one. This rectangle is called the **footprint**, and it is set in `config/config.yaml` as `bounding_box`. Everything outside it is ignored, because anything drawn there would land on the table rather than the model.
- **The projector.** Mounted above the model and **mirroring the laptop's screen**. This one fact shaped the whole design: anything on the laptop screen — buttons, the mouse pointer, notification pop-ups — would be projected onto the model. So the projected page shows map content only, and all controls live on a separate tablet (§9.2).
- **Access.** The rig stays set up between weekly sessions, so calibration (lining the image up with the model) mostly needs small corrections rather than starting over.
- **Users.** The team presents; visitors then use the tablet themselves.

### 2.3 How the plan was set

On 25 September the owner and Claude Code worked through the design one question at a time (a "grilling" session; recorded in `docs/DECISIONS.md`, 25 Sep). The outcomes that shaped everything since:

- One codebase (Pulse CBD), with the teammates' planning direction but rebuilt calculations.
- The footprint from the team's surveyed corners, replacing an earlier 5 × 4 km placeholder box.
- A projection with map content only; controls on a tablet, with a visitor view, a PIN-locked team panel and a statistics page; the tablet connects over the laptop's own Wi-Fi hotspot; the laptop keyboard as a backup.
- Congestion first from Transport for NSW's own label, then from a measure of our own, checked against it.
- Demand layers — live crowding, jobs, service gap, Opal 2016 — each switchable, on both travel zones and a grid, plus a combined pressure layer.
- Replays of recorded AM and PM peaks as the showcase default, with weekday peaks recorded automatically.
- No background map, because the model already has streets. Partly reversed on 4 November with an optional one that is off by default (§6.1).

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## 3. Key ideas and glossary

The rest of the report uses these terms. Each is explained here once.

### 3.1 Transport data

- **Transport for NSW (TfNSW).** The NSW government agency that runs Sydney's public transport. It publishes data through its **Open Data Hub**, which needs a free API key (a password-like code identifying who is asking for data). The project's key is in the file `.env` and is never committed to the code repository.
- **GTFS (General Transit Feed Specification).** The worldwide standard format for timetables. A GTFS "bundle" is a set of tables: **routes** (e.g. bus 333), **trips** (one run of a route, e.g. the 7:42 am 333 to

Bondi), **stops**, **stop times** (when each trip is scheduled at each stop), and **shapes** (the line on the map a trip follows). Pulse CBD downloads TfNSW's complete bundle, keeps only what touches the footprint, and stores it in a SQLite database (a single-file database). This is called the **static** timetable because it changes only when the timetable does.

- **GTFS-Realtime.** The companion live format. TfNSW publishes three kinds of live feed for each of seven modes (Sydney Trains, NSW TrainLink intercity, Metro, two light rail lines, buses, ferries):

    - **Vehicle positions:** where each vehicle is, which trip it is running, and optional extra fields including `congestion_level` (how congested the traffic around it is) and `occupancy_status` (how full it is).
    - **Trip updates:** predicted delays at upcoming stops — for example "this trip will reach its next stop 4 minutes late".
    - **Service alerts:** disruptions and notices.

  TfNSW adds its own extension (field number 1007) to report crowding per train carriage.

- **Delay.** In this project, a vehicle's delay is TfNSW's predicted delay at the vehicle's **next stop**, in seconds. Positive means late. It comes from trip updates, not from comparing the vehicle's position with the timetable ourselves.

- **Peak.** The busy commuting periods. Throughout the project, the **AM peak** is 7:00–9:00 and the **PM peak** 16:30–18:30, Sydney time, on weekdays.

### 3.2 Places and demand data

- **Travel zones (TZ21).** TfNSW divides NSW into small areas called travel zones (the 2021 version is TZ21) for its transport models. 405 of them touch the footprint. They vary greatly in size: tiny in the CBD, much larger in the suburbs.
- **TZP24 employment.** TfNSW's travel zone projections (2024 release) estimate how many jobs are located in each zone, by year. Pulse CBD uses the 2026 estimate. It counts people *working* in a zone, not living there, and it is a projection, not a census count.
- **Opal.** Sydney's ticketing smartcard. Passengers "tap on" when boarding and "tap off" when leaving. TfNSW released anonymised tap counts for one week in November 2016 (Opal Tap On and Tap Off, Release 2), by station for trains, light rail and ferries and by postcode for buses. That is the only public tap data at this level of detail, which is why such old data is used.
- **BOAM (Bus Opal Assignment Model).** A TfNSW dataset that combines Opal taps with bus tracking to estimate, for every bus trip at every stop, how many people were on board (in bands such as 21–40 passengers), the bus's capacity, and its scheduled and actual arrival times. It is published by day; the project uses 13 weekdays.
- **Postal areas (ABS POA).** The Australian Bureau of Statistics' postcode boundaries, used to place 2016 bus taps.

### 3.3 Methods and statistics

- **Map matching.** GPS positions wander off the road. Map matching moves ("snaps") each vehicle's reported position onto the nearest point of the line its trip follows, so it is drawn on the street, not beside it.
- **Segment.** A 200 m piece of a route's line. Street layers colour segments.
- **Pooling.** Combining readings from several vehicles. Here, every vehicle on the same street heading the same way in the last 10 minutes contributes to that street's colour, even if they are on different routes

(§7.5).

- **Level.** TfNSW's traffic and crowding fields are categories, not measurements. The project turns each into a number from 0 to 3 (0 = smooth or plenty of seats, 3 = severe or full) so readings can be averaged and then rounded back to a category.
- **Free-flow speed.** How fast buses move along a street when it is empty. The project estimates it from night-time recordings (§7.10).
- **Travel-time index (TTI).** Free-flow speed divided by current speed: how many times longer a trip takes now than when the street is empty. 1.0 means as fast as at night; 2.0 means twice as long.
- **85th percentile.** The value that 85% of readings are at or below. Used for free-flow speed because the very fastest readings may be errors, while the median would include slow readings caused by stops and lights.
- **Percentile rank.** Where a value sits among all values, from 0 (lowest) to 1 (highest). An area with more jobs than 90% of areas has a percentile rank of 0.9. Ranks have no units, so very different quantities (jobs, departures, taps) can be compared or combined.
- **Fifths (quintiles).** Sorting areas by a value and splitting them into five equal-sized groups. The colour then says "this area is in the top fifth", not "this area has more than X".
- **Cohen's weighted kappa (κ).** A score for how well two sets of categories agree, after removing the agreement expected by chance. 0 means no better than chance, 1 means perfect agreement; "weighted" means near-misses (one level apart) count partly. It matters because two measures that both say "smooth" most of the time will often match by coincidence.
- **Spearman's rank correlation (ρ).** Whether two measures rise and fall together, using their ranks; from −1 (opposite) through 0 (unrelated) to 1 (same order).
- **Homography.** The mathematical transform that maps a flat map onto a flat surface seen at an angle. Calibration computes one from at least four known points, so map coordinates land on the right spot of the model even if the projector isn't perfectly square to it.

### 3.4 Software terms

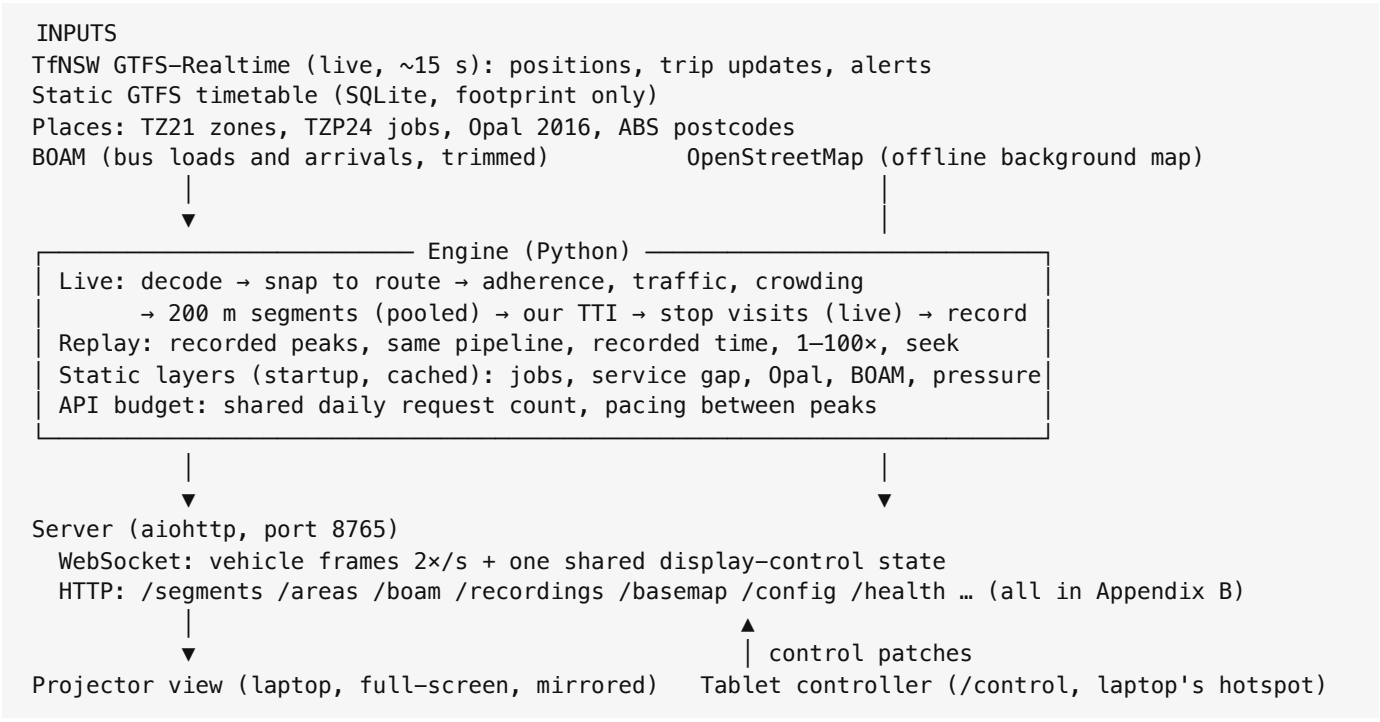
- **Server.** The Python program at the centre of the system, running on the laptop. It fetches the live data, does all the calculations, and serves the pages.
- **Engine.** The part of the server that processes data (decoding, map matching, calculations), as opposed to the part that talks to the pages.
- **Projector page** (`/`) and **tablet page** (`/control`). The two web pages the server provides. The projector page is shown full-screen on the laptop (and so on the model); the tablet page is opened on a tablet.
- **WebSocket.** A connection that stays open between a page and the server so either side can send messages at any time. The server sends vehicle positions about twice a second this way, and the tablet sends button presses back.
- **Display-control state.** One shared record, held by the server, of everything about what is shown: which layer is on, which mode is hidden, live or replay, and so on. Every page displays this one record (§4.2).
- **Replay / recording / session.** The server saves every live data download to disk (a recording). A **session** is a replayable window of recordings: an AM or PM peak on a given day, or the latest two hours.
- **launchd.** macOS's built-in service manager. Two launchd jobs keep the server running and record peaks on schedule.
- **API quota.** TfNSW limits how many requests one key may make per day — measured at about 60,000 (§5.1).

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# Part II — How it works

## 4. System architecture

### 4.1 The parts




There are three parts, all on one laptop:

1. **The engine** (Python, folder `engine/`). Every 15 seconds during peaks it downloads the live feeds, decodes them, works out which trip and route each vehicle is on, snaps it onto its route, classifies its punctuality, adds its traffic and crowding readings to the street it is on, computes our own speed-based congestion reading, notes when it passes a stop, and saves the raw download to disk. At startup it also builds the slower "static" layers — jobs, service gap, Opal, BOAM and the pressure model — in the background, and keeps them in a cache file so later startups are almost instant.
2. **The server** (`engine/publish/server.py`, using the aiohttp library, on port 8765). It sends the vehicle positions to every open page about twice a second over a WebSocket, answers requests for layers (streets, areas, BOAM, recordings, and so on), and holds the display-control state.
3. **Two web pages** (`renderer/`). The **projector page** (`index.html`) draws everything on an HTML canvas: background, area colours, street colours, route corridors and vehicles. The **tablet page** (`controller.html`) has the buttons, the legend and the statistics.

### 4.2 One shared display state

Every button on the tablet, and every key on the laptop keyboard, sends a small change (a "patch", e.g. "set the street layer to traffic") to the server. The server checks it is valid (`engine/publish/control.py`) and sends the complete updated state to every open page. Pages never change what is shown on their own; they only follow the server's state.

This is why the tablet and the projection can never disagree. With two tablets open, pressing a button on one updates the other. If a page reloads, it receives the current state and shows exactly what the others show. Everything except the alignment nudge resets to safe defaults when the server restarts, so the showcase always starts from a known state. All fields are listed in Appendix A.

### 4.3 Live and replay go through the same code

When replaying, the engine reads recorded downloads from disk in the order they were fetched and feeds them through exactly the same processing as live data, at the chosen speed. The engine treats each download's original fetch time as "now". This means the 10-minute windows used for street colours cover 10 *recorded* minutes even at 25× speed, and a replay looks exactly as the live view did at the time (with today's code). While a replay plays, live polling and the server's own recording pause (to save the data quota), so **after any demonstration, switch back to Live**.

### 4.4 The legend reads the engine's own numbers

Since 5 November the tablet's legend does not keep its own copy of thresholds. It reads them from the server (`/config` for punctuality and travel-time thresholds, `/areas?meta=1` for the value range of each fifth), so if a threshold is ever changed in `config/config.yaml`, the legend changes with it.

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## 5. Data sources

### 5.1 Live vehicle data (TfNSW GTFS-Realtime)

**What it gives:** vehicle positions, trip-update delays and service alerts for seven modes, refreshed about every 15 seconds; plus TfNSW's `congestion_level` and `occupancy_status` fields and per-carriage crowding for trains.

**Coverage, checked live on 23 October:** 164 of 269 buses reported a traffic level; crowding was reported for all buses and Metro and about half of Sydney Trains; light rail and ferries report neither field. Sydney Trains' traffic level is always "unknown" (it would be meaningless on rails). Since the teammate's fix merged on 30 October, about 70% of Sydney Trains show no usable delay (§7.2).

**The quota.** TfNSW doesn't document a daily limit, but on Sunday 25 October the key was refused after 59,964 requests counted from midnight and accepted again just after the next midnight. So the quota is about **60,000 requests per day, resetting at midnight Sydney time**. Polling all 15 endpoints every 15 seconds would cost about 86,000 a day — the server ran out at 7 pm that Sunday. Polling is now paced automatically (§12.3).

**Quirks the engine handles** (all discovered against the real feeds):

- Sydney Trains, Metro and CBD light rail use different identifiers in their live feeds than in the timetable, so their vehicles can't be joined to timetable trips by ID. They are identified by position instead: snapped to the nearest route line within 100 m.
- The ferry feed reports each vessel's current position and future scheduled ones together; only the freshest current one is kept.
- When a bus route is altered, its trip ID changes and stops matching the timetable; the engine falls back to matching on the route.
- Buses arrive as one statewide feed; only vehicles inside the footprint are kept.

### 5.2 Static timetable (TfNSW GTFS)

The complete GTFS bundle, filtered to the footprint and stored in SQLite (`data/static_cache/gtfs_cbd.db`). It was refreshed on 30 October because the August copy's calendar ended on 5 November, the day before the showcase. The current one runs **30 October 2026 – 28 January 2027** and has 546 routes, 2,905 stops and 3,135 route shapes in the footprint. The old copy is kept as `gtfs_cbd.2026-08-19.db`. It provides route names

and colours, stop locations, route lines for map matching and drawing, and scheduled departures for the service gap.

### 5.3 Travel zones and jobs (TZ21, TZP24)

Zone boundaries and 2026 job estimates, taken from the team's branch (stored with Git LFS, a way of keeping large files alongside code). `scripts/prepare_demand_data.py` cuts the 109 MB statewide zone file down to the footprint. 405 zones touch the footprint. Caveat: zones differ hugely in size, so a large suburban zone can have many jobs simply because it is large. That is one reason the 150 m grid exists as an alternative (§7.6).

### 5.4 Opal 2016

From data.nsw.gov.au: tap-ons and tap-offs in 15-minute bins for 21–27 November 2016. Trains, light rail and ferries are given by station name; buses only by postcode. Caveats: it is ten years old and predates the Metro and the CBD light rail, so those lines show nothing.

### 5.5 Postcode boundaries (ABS Postal Areas 2021)

Used only to place 2016 bus taps. 40 postcodes touch the footprint.

### 5.6 BOAM

From TfNSW's Open Data Hub (needs a login), imported with `scripts/import_boam.py`, which keeps only footprint rows (about 2.7 MB per day instead of about 400 MB). **13 weekdays are used**: 28–30 September (school holidays) and 12–16 and 19–23 October (school term). Four weekend days are kept but not averaged. Caveats: since March 2026 BOAM's "actual" arrival times are themselves TfNSW's live predictions, so BOAM isn't fully independent of the live feed; people who don't tap (notably some students) aren't counted; and three of the 13 days are school holidays (loads match term days to about one percentage point, so they were kept).

### 5.7 OpenStreetMap (background map)

The community-made world map, downloaded once on 4 November through the Overpass service by `scripts/fetch_basemap.py`, and stored offline in `data/basemap/basemap.json` (about 3 MB): roads, rail, water, parks, coastline and 753 place and street names. It is credited "© OpenStreetMap contributors" on the tablet.

### 5.8 Our own recordings

Every live download is saved, trimmed to a footprint padded by 10% (so vehicles have some history before entering the frame; untrimmed statewide feeds cost about 14 GB in a few hours). They have been recorded continuously since 24 October, at about 3.5 GB per day, 27 GB in total. Recordings from August and September were deleted: the old setup was being rate-limited and captured only 1–3 bus updates an hour. Only two weekend sessions from then (22 August, 6 September) are still usable.

### 5.9 Stop visits

The teammate's stop history (merged 30 October) records one row each time a vehicle passes a stop, with its crowding and delay at that moment. There were 865,907 visits by 4:27 pm on 5 November, recorded live since 30 October. Replays don't add visits (§7.13).

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## 6. What it shows and how it is controlled

### 6.1 On the projection

**Vehicles** are always shown. Each is snapped to its route, moved smoothly between updates, and drawn as an arrow pointing the way it is travelling, with a short tail. Rail vehicles (trains, metro, light rail) are drawn larger than buses and ferries. Each of five groups — trains, metro, light rail, buses, ferries — can be hidden. The arrow colour follows the chosen **lens** (what the vehicle colour shows):

- **Schedule** (default): punctuality from trip-update delays.
- **Traffic:** TfNSW's congestion level for the vehicle (buses only).
- **Crowding:** TfNSW's occupancy for the vehicle.

While an area layer is on, vehicles turn white so they don't compete with the area colours.

**Street layers** colour the streets themselves. Only one is shown at a time:

| Layer | What it shows | Source | Time basis |
|---|---|---|---|
| Traffic | How congested TfNSW says the traffic around buses is | TfNSW `congestion_level` | Live, last 10 min |
| Traffic (our measure) | How much slower buses are moving than on the same street at night | Our travel-time index (§7.10) | Live, last 10 min |
| Crowding | How full vehicles are | TfNSW `occupancy_status` | Live, last 10 min |
| Typical bus load | How full buses typically are leaving each stop | BOAM | Average weekday AM or PM peak |
| Where buses lose time | How much further behind timetable buses typically fall per 100 m | BOAM | Average weekday AM or PM peak |

Every street layer can show the two directions of a street in four ways (tablet "Street directions", or **D** on the keyboard):

- **One colour** (default): both directions combined into one colour per street.
- **Side by side:** each direction drawn as its own line on its own side of the street, like traffic colours on navigation apps.
- **To city / From city:** only traffic heading towards, or away from, Town Hall.

A street whose colour rests on only one vehicle is drawn **faint** (35% opacity), as a hint rather than a finding. Each stretch of street has exactly one line per direction shown, so no colour ever peeks out from under another (§7.12).

**Area layers** shade the model in patches. Only one is shown at a time, on either **travel zones** or a **150 m grid** (switchable, so the team can judge on the real model which reads better): Jobs, Service gap, Opal 2016, and Pressure (combined, updated live).

**Stops and routes** (teammate's work, merged 30 Oct):

- **Stop markers** for all stops (off by default; it is a lot of dots).
- **Stop history:** each stop coloured by its typical delay (median or mean) or how often vehicles arrive crowded, over everything recorded live. A stop needs at least 20 visits to be coloured.

- **Route filter:** single out up to eight routes by name (e.g. 333, T1, L2). Their vehicles, line and stops stand out in fixed colours and everything else fades back. Each route keeps its colour when another is removed.

**Background map** (added 4 Nov): **Plain** (default; the model already has streets, and anything drawn would compete with them), **Detailed** (faint roads, rail, water and parks from OpenStreetMap) or **Full** (Detailed plus suburb, station, landmark, park, water and street names, placed so they never overlap). It is mainly useful on a flat screen or laptop.

**Keeping the projection clean.** There are no buttons, hints or mouse pointer on the projected page, and everything is clipped to the footprint. An optional **status line** (team panel, or **H**) shows the mode, clock and replay state. It is currently the only way the projection itself says "this is a replay", so either turn it on during replays or say so out loud.

### 6.2 Time: live and replay

- The system starts in **Live**.
- **Replayable sessions:** every recorded AM peak (7:00–9:00) and PM peak (16:30–18:30) that has at least 10% of the expected bus updates, plus "Latest 2 hours". The 10% minimum hides sessions so sparse they would replay as an almost empty model.
- **Choosing one (since 5 Nov).** The tablet's Time section has three buttons of the same style: **Live now**, **Latest 2 hours** and **Replay mode**. Replay mode opens a **day dropdown** listing every recorded day, newest first (14 days, 25 peaks today), next to **AM peak** and **PM peak** buttons. Opening it changes nothing on the model until a peak is tapped. A missing peak's button is greyed out, and a note flags peaks under 90% recorded (e.g. "partly recorded: AM 66%, PM 66%" for the weekend of 31 Oct–1 Nov). Changing the day while a peak is playing jumps to the same peak on the new day. Only one Time button is lit at a time. Before 5 November the tablet showed only the first seven sessions (the latest plus three days), even though the server listed all of them.
- **Speed:** 1× (real time), 10×, 25× (default; a two-hour peak in under five minutes) or 50× on the tablet; the server accepts 1–100×. Replays loop.
- **Pause and play** (tablet, or Space): the projection freezes and resumes without jumps.
- **Scrubbing:** tap anywhere on the timeline, use the –5/+5 minute buttons, or ←/→ for one minute. On a jump, the engine first processes the 10 recorded minutes before the target as fast as it can, so the street layers are already filled when playback resumes.
- **Remember:** live polling and recording pause during any replay. Switch back to Live afterwards.

### 6.3 The tablet (`/control`)

The tablet has three tabs.

**Display** (for visitors) holds:

- time and replay controls;
- which vehicle groups are shown, and the dot colour (lens);
- the street layer and street directions;
- stop markers and stop history (median or mean);
- the route filter;
- the area layer and area shape (zones or grid), and the BOAM peak (AM or PM);
- the background map;
- **What the colours mean**, a legend for whatever is currently on.

Since 5 November the legend gives every colour a definition with its cut-off, not just a name. For example:

- "On time: up to 5 min trains, buses & ferries; 6 min intercity trains; 2 min metro & light rail behind the timetable."
- "Jobs, second fifth: 14–28 jobs per 150 m square."

Where TfNSW only provides categories (its traffic and crowding fields), the legend says so and names the feed value behind each colour rather than inventing ranges. It also explains the street rounding rule, solid versus faint lines, and the stop-history bands. Every definition is in §8.

**Team** (behind a PIN, `3704`; the check happens in the browser only, so it keeps visitors out but isn't real security) holds:

- the theme (dark or light);
- the verification overlay (route corridors drawn bright, to check alignment);
- the status line on the projection;
- alignment: the nudge (shift by 2 px or 20 px, and scale), **corner fit** (drag the footprint's four corners onto the model's; teammate's, 5 Nov) and crosshair calibration;
- pressure weights (0–3 for each input);
- system health: vehicles per mode, how old each feed is, API requests used today, and the current polling interval.

**Stats** shows on-time percentages by mode over the last 5, 15 and 60 minutes, active service alerts affecting the footprint, and the busiest interchanges.

### 6.4 Laptop keyboard (backup)

Every key goes through the server, so the tablet stays in sync:

C calibrate · F corner fit · V verification overlay · H status line · T theme · M background map · S stop markers · O stop history · L dot colour · K street layer · A area layer · G zones/grid · B BOAM peak · D street directions · Space pause/play · 1–5 mode groups · arrow keys nudge (Shift ×10) · [ ] scale · 0 reset nudge.

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## 7. How every number is calculated

This section explains each calculation in enough detail to reproduce it, with the reason behind each choice and a worked example. File names point to where the code lives.

### 7.1 Putting vehicles on their routes (`engine/normalise/map_match.py`, `enrich.py`)

**The problem.** A live feed gives each vehicle a GPS position and, usually, a trip ID. GPS positions drift several metres off the road, and some modes' trip IDs don't match the timetable at all.

**What the engine does**, for every vehicle on every update:

1. **Identify the trip and route.** If the live trip ID is in the timetable, use it. If not, but the route ID is, use the route. If neither matches, find the nearest route line within 100 m and use that. This last step is what places Sydney Trains, Metro and CBD light rail, whose live IDs use a different naming scheme from the timetable. Each vehicle records which of these worked, so nothing is silently guessed.
2. **Snap to the line.** Move the position to the nearest point on the trip's route line, and record how far along the line that point is (its "distance along shape"). Drawing uses the snapped position, so vehicles sit on

the street; the distance along the line is what our speed measure (§7.10) and the street segments (§7.5) use.

**Why it was rebuilt for speed.** At first the search compared every vehicle with every whole route line. Almost every route line in the timetable passes through the CBD, so the search could rule almost none out: it took about 14 ms per vehicle and 92% of the server's processor time. The search index now holds 32-point pieces of each line, so most are ruled out instantly. The live server went from about 85% processor use to a few percent, and replays from about 11× to about 96× real time.

### 7.2 Punctuality (`engine/normalise/adherence.py`, `trip_updates.py`)

**The measure.** A vehicle's delay is TfNSW's predicted delay, in seconds, at the vehicle's **next stop**, taken from the trip-update feed. Comparing with the next stop (not the trip's first stop, and not the vehicle's position against the timetable) means the colour describes how late the vehicle is *now*, where it is.

**The threshold.** Each mode has an on-time threshold $T$ (`config/config.yaml`, `adherence_thresholds_seconds`):

| Mode | $T$ | Late | Very late |
|---|---|---|---|
| Sydney Trains, buses, ferries | 5 min (300 s) | 5–10 min | over 10 min |
| NSW TrainLink intercity | 6 min (360 s) | 6–12 min | over 12 min |
| Metro, light rail | 2 min (120 s) | 2–4 min | over 4 min |

The rules:

| State | Rule | Colour |
|---|---|---|
| On time | delay ≤ $T$, including running early | green |
| Late | $T <$ delay ≤ $2T$ | amber |
| Very late | delay > $2T$ | red |
| Cancelled | TfNSW marks the trip cancelled | red |
| Unknown ("no schedule data") | no usable prediction | grey |

**Worked example.** A bus predicted to reach its next stop 7 minutes late: $T$ = 5 min, $2T$ = 10 min, so it is "late" (amber). A metro 3 minutes late is also "late", because metro's $T$ is 2 minutes; at 5 minutes late it would be "very late".

**Where the thresholds come from.** The rail, metro and light rail values came from the original project brief as sensible defaults resembling TfNSW's published standards; they have not been checked against an official source. Buses and ferries were set to 5 minutes by analogy with suburban rail. The "twice the threshold" rule for very late is the project's own choice. Running early counts as on time, because the brief's states don't include "early". In the final report these should be called *the project's simplified thresholds*.

**The placeholder-zero fix (teammate's, merged 30 Oct).** Some feeds send a delay of exactly 0 seconds for every stop of a trip when they have no real prediction. Treating that as "on time" made almost every Sydney Train green. Now an exact 0 s across a whole trip counts as "no prediction", so about 70% of Sydney Trains show grey. The same fix makes Metro, light rail, ferries and NSW TrainLink read the next stop's delay rather than the trip's first stop's (which was always 0). On-time percentages on the Stats tab changed accordingly.

### 7.3 Interchange activity (`engine/normalise/aggregate.py`)

For each of nine interchanges (Central, Town Hall, Wynyard, Martin Place, Circular Quay, Barangaroo, Gadigal, Museum, St James), the engine counts the **distinct** vehicles seen within **300 m** in the last **5 minutes**. "Distinct" matters: a train waiting at a platform across several updates counts once, not once per update. The counts feed the "busiest interchanges" bars on the tablet's Stats tab.

Until 5 November they also drove glowing rings on the projection that grew and brightened with activity (the "interchange pulses"). The owner had them removed: they are no longer drawn, and their toggle and **P** key are gone.

### 7.4 TfNSW's traffic and crowding fields (`engine/normalise/levels.py`)

TfNSW's two "how is it going" fields are categories. The project turns each category into a level from 0 to 3 so readings from several vehicles can be averaged:

| Level | Traffic (`congestion_level`) | Crowding (`occupancy_status`) |
|---|---|---|
| 0 | RUNNING_SMOOTHLY | EMPTY, MANY_SEATS_AVAILABLE |
| 1 | STOP_AND_GO | FEW_SEATS_AVAILABLE |
| 2 | CONGESTION | STANDING_ROOM_ONLY |
| 3 | SEVERE_CONGESTION | CRUSHED_STANDING_ROOM_ONLY, FULL, NOT_ACCEPTING_PASSENGERS |
| none | UNKNOWN or missing | missing |

Some notes on these fields:

- **Seven crowding categories become four.** EMPTY is folded into "many seats" and the three "no room" states into one, because seven colours can't be told apart on a projection.
- **Traffic counts buses only.** Metro reports a level, but it isn't road traffic. Sydney Trains' level is always unknown.
- **Trains without a vehicle-level crowding value** use the **middle carriage** of the per-carriage readings.
- **There are no numbers behind these categories.** TfNSW doesn't publish how its traffic levels are decided (from speed, delay or something else) or how many passengers each crowding category means. So no numeric range can honestly be given for them. That is exactly why the project built its own congestion measure with stated cut-offs (§7.10).

### 7.5 Street segments and pooling (`engine/normalise/segments.py`)

**Cutting streets into pieces.** Each route's line is cut into **200 m pieces** by distance along it. A vehicle's reading belongs to the piece it is on.

**The problem pooling solves.** The pieces belong to routes, not to streets. A street served by ten routes therefore had ten overlapping sets of pieces, each with its own handful of buses. At the 8:00 am peak on 4 November, **about 80% of coloured pieces rested on a single vehicle** — 79% within 1.5 km of Town Hall and 84% beyond 4 km. They were drawn worst on top, so a busy street showed its single worst bus: a level worse than the average of all its buses at 28% of places where pieces overlapped. A concern was raised that one bus was standing for a whole street, and the measurements confirmed it.

**What pooling does.** Each piece's value now combines every vehicle, from any route, seen in the last **10 minutes** on pieces of the same street heading the same way. Two pieces count as "the same street, same way" when:

- their midpoints are within **100 m** of each other along the street and **25 m** across it (parallel CBD streets are about 100 m apart, so 25 m keeps them separate);
- their headings are within **30°**;
- the check works from both pieces' points of view, so the result doesn't depend on which piece is checked first.

Each vehicle counts once, at its latest reading. The two directions stay separate, because inbound and outbound traffic often differ.

**What each piece carries:**

- the mean level for each layer;
- **how many vehicles** that mean rests on (used to draw thin evidence faint);
- the same values pooled over **both directions** (used by "One colour");
- whether the piece heads **towards Town Hall** (used by "To city" and "From city").

**Turning the mean back into a colour.** The mean is rounded to the nearest level:

| Mean | Level |
|---|---|
| under 0.5 | 0 |
| 0.5 to under 1.5 | 1 |
| 1.5 to under 2.5 | 2 |
| 2.5 and over | 3 |

*Worked example:* three buses on a street report "running smoothly", "running smoothly" and "congestion". Their levels are 0, 0 and 2, so the mean is 0.67, which rounds to level 1, "stop and go". With three vehicles it is drawn solid; if only the congested bus had been there, it would be level 2 but drawn faint.

**Result.** At the same 8:00 am peak, pieces resting on one vehicle fell to **11% in the CBD, 20% at 1.5–4 km and 32% beyond 4 km**. So faint streets now cluster where data really is thin, and a solid colour means at least two vehicles agree.

**Speed.** Which pieces count as the same street never changes, so it is worked out once at startup (about 110 ms). After that a full snapshot of all streets takes about 40 ms at peak; the first pooled version took 1.4 s.

### 7.6 Jobs, departures and the service gap (`engine/demand/areas.py`)

These layers use two kinds of area:

- **Travel zones** (405 in the footprint);
- a **150 m grid**, whose squares are all the same size, so large zones don't stand out just for being large.

Grid squares take an area-weighted share of the zones they overlap. For example, a square lying half in a zone with 1,000 jobs, where that half is 5% of the zone's area, gets 50 jobs from it.

**Jobs:** the zone's TZP24 2026 job estimate.

**Departures:** a measure of how much public transport a worker in the area can reach. It counts scheduled departures, of any mode, from stops within **200 m** of the area, between **7:00 and 9:00** on a reference weekday (**Tuesday 10 November 2026**, inside the current timetable). A stop counts for every area within 200 m of it, and a trip counts once for each stop it serves there. So it measures service you could walk to, not the number of distinct trips.

**Service gap:** the area's percentile rank for jobs minus its percentile rank for departures, among areas with any jobs.

- It ranges from –1 to +1.
- **Positive** means more jobs than service compared with the rest of the footprint (relatively under-served).
- **Negative** means more service than jobs.

*Worked example:* an area with more jobs than 90% of areas (rank 0.9) but more departures than only 30% (rank 0.3) has a gap of +0.6. That is "much more jobs than service".

**Why a rank difference and not a ratio.** Jobs divided by departures explodes when departures are near zero, and comparing the two raw numbers would mix units. Ranks put both on the same 0–1 scale with no units.

**Bands:**

| Gap | Label |
|---|---|
| –0.40 or less | much more service than jobs |
| –0.40 to –0.15 | more service |
| –0.15 to +0.15 | balanced |
| +0.15 to +0.40 | more jobs |
| over +0.40 | much more jobs than service |

**Fifths (jobs and Opal layers).** Areas with any value are ranked and split into five equal-sized groups. Areas with none aren't coloured. The actual value range of each fifth is in §8.3. The ranges differ between zones and the grid, and the tablet shows the ones for whichever shape is on.

### 7.7 Opal 2016 (`engine/demand/areas.py`)

The layer shows the average weekday of 21–25 November 2016: tap-ons plus tap-offs between 7:00 and 9:00.

**Stations** (train, light rail, ferry) are placed by matching their names to today's stops. For example, "Circular Quay, No. 3 Wharf" becomes the place "Circular Quay" plus the stop type "Wharf".

**Bus taps** are given only by postcode, so they are spread out:

- each postcode's bus taps are scaled by the share of its area inside the footprint;
- the result is split evenly across today's bus stops in that postcode.

Each area then sums the taps placed inside it.

### 7.8 Pressure (`engine/demand/pressure.py`)

Pressure is one score per area combining five inputs. Two are live; three come from fixed data:

| Input | Kind |
|---|---|
| Traffic now | live |
| Crowding now | live |
| Service gap | static |
| Opal 2016 | static |
| Typical bus load (BOAM, chosen peak) | static |

**How the score is built:**

1. **Turn each input into a percentile rank** (0 to 1) across the areas that have it. Ranks are robust to outliers: one huge station doesn't squash everyone else into the bottom colour.
2. **Fill gaps with the neutral 0.5.** Areas with no live vehicles right now, or no BOAM sections, get 0.5 for that input. Simply dropping missing inputs had pushed empty areas to the top of the ranking.
3. **Combine with weights:** score = $\Sigma\ w_i \cdot r_i \div \Sigma\ w_i$, where each weight $w_i$ is set from 0 to 3 on the team panel (default 1).
4. **Colour by fifths of the scores.** Averaged ranks bunch around 0.5, so fixed bands would put nearly everything in the middle colour.

Jobs are left out because they are already inside the service gap. Counting them twice would overweight them.

The score is **relative only**: it says which areas are under the most pressure compared with the rest of the footprint, not against any planning standard. It updates every few seconds as the live inputs change.

### 7.9 BOAM bus sections (`engine/demand/boam.py`)

A **section** is the stretch between two consecutive stops, a → b, on a bus trip. For each section, each peak (AM 7:00–9:00, PM 16:30–18:30) and weekdays only, two values are worked out.

**Load: how full the bus is leaving stop a.**

- BOAM gives passengers on board in bands (0–20, 21–40, …). The middle of the band is used: 21–40 becomes 30.5.
- That number is divided by the bus's **total capacity**, then averaged over every weekday trip on the section in the peak.
- *Example:* a band of 41–60 on a bus with capacity 75 gives 50.5 ÷ 75 = 67%, in the top band.
- This assumes BOAM's occupancy at a stop is the load departing it; BOAM's documentation doesn't say.

**Time lost per 100 m: how much further behind timetable the bus falls on this section.**

- Time lost per 100 m = ((actual_b – scheduled_b) – (actual_a – scheduled_a)) ÷ section length × 100.
- *Example:* a bus 60 s late at a and 120 s late at b, 400 m apart, lost 60 s over 400 m, which is 15 s per 100 m.
- Zero or negative means the bus kept to time or caught up.
- Measuring the *change* in delay between two stops, rather than the delay itself, means a bus that started late elsewhere doesn't make this section look slow.

**Data rules:**

- A change in delay of more than 15 minutes between two stops is treated as a data error and ignored.
- A section needs **at least 3 trips** in the period to be shown.
- Sections are joined to the timetable by **stop ID** (100% match). BOAM's trip IDs don't match the timetable's, but its stop IDs do.

**Long sections.** Sections over **800 m** are left out of the time-lost layer but kept in the load layer.

- These are express routes over motorways and bridges, up to 4.9 km long. They are 5% of sections but 24% of the coloured length.
- An average delay over several kilometres can't show *where* the time was lost.
- Load really is constant along a section, because nobody boards or alights between stops, so the load layer keeps them.

**Bands** (absolute, so days added later stay comparable):

- **Load:** ≤ 15%, 15–30%, 30–45%, 45–60%, > 60% of capacity. Averaged over a two-hour peak, section loads rarely pass 50%, so 20%-wide bands left nearly every street in the bottom colour.
- **Time lost:** ≤ 0, 0–3, 3–8, 8–15, > 15 s per 100 m.

**Both directions.** For "One colour", each section also gets a value pooled, weighted by trip count, over every section whose midpoint lies within 25 m of its line in either direction.

### 7.10 Our own congestion measure: the travel-time index (`engine/normalise/travel_time.py`)

**Why build one.** TfNSW's traffic label is published without a method (§7.4). A measure we can explain lets the project say exactly what a colour means, and test whether TfNSW's label behaves like a measure of slow streets (§10).

**The reading.**

- For each bus placed on its route line, the engine waits until the bus has moved **at least 300 m** along the line since the last reading.
- It then computes the bus's average speed over that stretch. Stops and traffic lights are included on purpose, because the free-flow speed below includes them too, so they cancel out.
- The reading is attributed to the 200 m piece at the stretch's midpoint.

**The free-flow speed.** This is a street's "empty" speed: the **85th percentile** of the same kind of reading taken **at night (22:00–05:00)** from our own recordings, for each 200 m piece.

- Pieces with too few night readings fall back to the whole route's night speed.
- The current baseline (`scripts/build_freeflow.py`) has 4,712 pieces from 7 nights, plus 217 whole-route fallbacks.
- Pieces with no baseline at all show grey.

**The index.** TTI = free-flow speed ÷ current speed, capped at 10.

- *Example:* a street buses cover at 30 km/h at night, now crossed at 12 km/h, gives TTI 30 ÷ 12 = 2.5. The trip takes two and a half times as long as at night.

**Levels on the model** (`derived_congestion.thresholds_tti`), chosen to be easy to explain to visitors:

| TTI | Level |
|---|---|
| up to 1.5 | running smoothly |
| 1.5–2 | stop and go |
| 2–3 | congested |
| over 3 | severely congested |

For the statistical comparison with TfNSW, different cut-offs are used (2.66 / 2.98 / 3.32). These were fitted so our four levels occur as often as TfNSW's do. Otherwise the comparison would mainly measure a difference in how often each says "congested" (§10).

**Why a night baseline, and not the timetable?** Comparing with the timetable was tried first and failed: timetables are padded, so buses "gained" time even in severe congestion (§10.1).

### 7.11 Street colours from BOAM and from the live layers together

The live street layers (Traffic, Traffic (our measure), Crowding) use §7.5. The BOAM layers (Typical bus load, Where buses lose time) use §7.9. They answer different questions:

- **Live layers:** what is happening right now, or during the replayed moment.
- **BOAM layers:** what a typical weekday peak looks like, averaged over 13 days.

Both obey the same direction modes and the same drawing rules (§7.12).

### 7.12 How street lines are drawn (`renderer/index.html`, `buildStreetCanvas`)

**Line style.** Each street gets one solid line, 4.5 pixels wide at 95% opacity. Streets resting on one vehicle are drawn at 35% opacity. "Side by side" uses two 4 px lines, each shifted 2 px to its own side, so the pair just touch.

**No glow (removed 4 November).** Lines used to have a soft 9 px halo underneath. On shared streets one halo was drawn per route, so busy streets glowed brighter even at the same level. The halo also hid small calibration errors, so alignment needs checking on the model again.

**One line per stretch.** Before 4 November, overlapping lines were stacked, so a red line on top could leave green peeking out at its edges. Of 1,882 such clashes at the 8:00 am peak on 4 November:

- 72% were a street's two directions drawn on top of each other;
- 28% were pieces of different routes slightly offset along the same street.

Now lines are laid down best evidence first:

1. solid before faint;
2. then more vehicles (or more BOAM trips);
3. then the worse level.

A later line draws only the parts of itself that no earlier line already runs *beside*. The check samples each line every 2 px and looks within 5 px across and ±1 px along. It counts only lines going the same way, or either way in "One colour".

"Beside", not "near", matters. The next piece of the same route starts where the previous one ended, and it must join on without leaving a gap.

**Performance.** All street lines are drawn once into a stored image whenever the data, layer, direction mode or calibration changes, rather than every frame.

### 7.13 Stop history (`engine/history/`, teammate's work)

**What a visit is.** A visit is recorded when a trip's "next stop" changes, meaning the vehicle has just passed that stop. The reading kept is the **last** crowding and delay seen while that stop was still next: what the vehicle looked like as it left. This uses TfNSW's own next-stop value (present on 97.4% of trip updates, checked live), rather than guessing from distance to the stop.

**What is computed per stop,** over everything recorded:

- median and mean delay;
- the share of arrivals at crowding level 1 or more ("few seats" or fuller);
- delay split by whether vehicles arrived seated or crowded. Each side needs at least 5 visits, otherwise no figure is given rather than one based on two readings.

Delays outside –30 minutes to +3 hours are treated as glitches and ignored.

**Median versus mean.** Both are offered because they say different things. The median is the typical vehicle; the mean also carries how bad the occasional very late one is. A big gap between them marks an unreliable stop.

**What a stop's colour does and doesn't say.** A vehicle arrives at a stop carrying all the delay it picked up earlier in its trip. A red stop therefore means "vehicles are typically late by the time they get here", not "this stop causes delay". That is also why the CBD tends to look greener: it has more rail, and buses pick up more delay towards the ends of their routes. To find where delay is *added*, compare consecutive stops, or use the BOAM "Where buses lose time" layer (§7.9).

**A finding from the teammate's notes (as of 4 November).** At 610 of 662 stops (92%) with enough visits, crowded vehicles arrived later than seated ones. That is an association, not proof that crowding causes delay: crowded vehicles are likely also the ones running at peak times and on the busiest routes, which are late for other reasons too.

**Only live data is recorded (fixed 30 October).** The visit recorder sat in code that replays also run through, so every replay, and every loop of a replay, wrote recorded visits into the history again. Now visits are written only while live. Checked: 30 seconds of a 50× replay added no visits.

**Speed.** The `/stop_history` address recomputes everything at most once a minute, without holding up anything else on the server. A full recompute takes about 2 seconds at 866,000 visits (5 November). A cached answer takes 0.03 seconds.

### 7.14 Background map (`scripts/fetch_basemap.py`)

**Downloading.** The map data comes from OpenStreetMap, through its public Overpass service. The public servers failed several times (errors 502 and 504), so the script retries, falls back to a second server, and keeps the raw download so it can be rebuilt without downloading again (`--cached`).

**Drawing.** The map is drawn as lines and shapes through the same projection as everything else, so it lines up with the model after calibration. Picture tiles, the usual way web maps work, would have needed warping to match.

**What each option shows:**

- **Plain** (the default) draws no map at all.
- **Detailed** draws major and minor roads, rail, water, parks and coastline. These are kept faint so data stays on top, and hidden while the verification overlay is on. Two problems found while checking it:
    - Road tunnels are left out, because drawing the Harbour Tunnel on the surface would mislead on a physical model.
    - Each water and park outline is filled separately. Filled together, overlapping outlines cancelled each other out and the harbour vanished.
- **Full** adds 753 labels. In order of importance: suburbs, water, stations and wharves, landmarks, parks over 4 hectares, neighbourhoods, then street names.
    - There is one label per wharf ("Circular Quay", not one per berth).
    - Street names are angled along the road and repeated about every 1 km.
    - Labels are placed greedily in that order. Any label that would overlap one already placed, or wouldn't fit wholly on the model, is skipped.
    - Bus interchanges and railway junctions aren't labelled, because they crowded out stations and places.

- Labels are drawn above the street colours and below the vehicles.
- At projector resolution the smallest labels (10 px street names) may be too small to read, so Full is probably most useful on a screen.

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## 8. Every colour, defined

This is what the tablet's *What the colours mean* section states (since 5 November), in one place. **Absolute** bands mean the same thing on any day. **Relative** bands rank areas against each other, so they say "top fifth of the footprint", not a fixed amount. **Categorical** bands are TfNSW's own categories, which have no published numbers behind them.

### 8.1 Vehicles (the lens)

| Lens | Colour | Definition |
|---|---|---|
| On schedule? | Green — on time | Up to 5 min behind the timetable (trains, buses, ferries), 6 min (intercity), 2 min (metro, light rail). Early counts as on time |
| | Amber — late | 5–10 min, 6–12 min, 2–4 min late respectively |
| | Red — very late / cancelled | Over 10, 12 or 4 min late, or the trip is cancelled |
| | Grey — no schedule data | No live delay prediction for the vehicle's trip |
| Traffic (TfNSW) | Green / yellow / orange / red | The bus's `congestion_level`: RUNNING_SMOOTHLY / STOP_AND_GO / CONGESTION / SEVERE_CONGESTION. TfNSW publishes no numbers behind these |
| | Grey | UNKNOWN or missing. Always the case for trains and metro, which aren't in road traffic |
| Crowding (TfNSW) | Blue / purple / pink / red | EMPTY or MANY_SEATS_AVAILABLE / FEW_SEATS_AVAILABLE / STANDING_ROOM_ONLY / CRUSHED_STANDING_ROOM_ONLY, FULL or NOT_ACCEPTING_PASSENGERS. These are categories, not passenger counts. Trains use the middle carriage |
| | Grey | No occupancy sent |

### 8.2 Streets

The live layers use the rounding rule in §7.5. A solid line rests on 2 or more vehicles; a faint line on only 1.

| Layer | Bands | Kind |
|---|---|---|
| Traffic, Crowding (TfNSW) | Levels 0–3 as in §8.1, averaged per street over the last 10 min and rounded | Categorical |
| Traffic (our measure) | Takes up to 1.5×, 1.5–2×, 2–3×, over 3× as long as the same street at night. Grey: no night baseline for that street | Absolute |
| Typical bus load (BOAM) | 15% or less, 15–30%, 30–45%, 45–60%, over 60% of the bus's total capacity | Absolute |
| Where buses lose time (BOAM) | 0 or less (held or gained time), 0–3, 3–8, 8–15, over 15 s lost per 100 m. Over 15 s means more than a minute lost on a 400 m stretch | Absolute |

### 8.3 Areas

These are the actual ranges as built on 5 November. The tablet reads them live from `/areas?meta=1`. The bottom fifth reads "up to", because small fractions of a job round down to 0.

| Layer | Lowest fifth | 2nd | Middle | 4th | Highest fifth |
|---|---|---|---|---|---|
| Jobs per travel zone | up to 391 | 393–726 | 727–1,260 | 1,263–2,983 | 3,065–21,409 |
| Jobs per 150 m square | up to 14 | 14–28 | 28–54 | 54–128 | 128–11,761 |
| Opal taps per travel zone | up to 114 | 114–210 | 212–435 | 444–630 | 638–48,327 |
| Opal taps per 150 m square | up to 23 | 23–44 | 47–92 | 93–186 | 192–45,806 |

All four are relative. Note how uneven the top fifth is: it runs from about 3,000 to over 21,000 jobs per zone. The busiest places, such as the CBD core and the big stations, are far above everything else, which is why fifths are used rather than evenly spaced bands.

- **Service gap** (fixed bands on the rank difference, §7.6): –0.40 or less, much more service than jobs; –0.40 to –0.15, more service; –0.15 to +0.15, balanced; +0.15 to +0.40, more jobs; over +0.40, much more jobs than service.
- **Pressure:** fifths of the combined score (§7.8). It has no fixed values; only the order matters.

### 8.4 Stops and background

| Overlay | Bands |
|---|---|
| Stop history, delay (median or mean on arrival) | Under 1 min, 1 to under 2 min, 2 to under 4 min, 4 min or more late. Early counts as under 1 min |
| Stop history, when the crowding lens is on | Share of arrivals with few seats or fuller: under 10%, 10 to under 25%, 25 to under 50%, 50% or more |
| Stop history, any | Stops with fewer than 20 visits aren't coloured |
| Background map | Not data: roads, rail, water and parks, drawn faint |

---

# Part III — Why it is built this way

## 9. Design decisions

### 9.1 Why the teammates' calculations weren't reused

The teammates' dashboard set the project's direction, but its congestion score had flaws that would have made the model misleading:

| Flaw in the dashboard's congestion score | Why it misleads |
|---|---|
| A bus free-flow speed (30 km/h) applied to trains and Metro | Rail isn't in road traffic, so "congestion" on rail meant nothing |
| Absolute delay treated as local | A bus that left Parramatta late reads as congested in the CBD, where it may be moving freely |
| Instantaneous speed | A bus stopped at a stop or red light reads as slow |
| Bunching that ignores the timetable | High-frequency routes run close together by design |
| Grid cells averaging whatever is in them at that instant | No history, and results depend on arbitrary cell boundaries |

Its "demand" was jobs per zone. That describes land use, not transit demand, so here it is labelled **jobs** and used inside the service gap. The dashboard also had a mapping-service API key committed in plain text.

## 9.2 The major decisions, explained

**The display state lives on the server and is broadcast.**

- *Why:* the projector mirrors the laptop, so controls must be elsewhere, and two places (tablet and keyboard) can change the display. With one copy on the server, every page shows the same thing, and a reloaded page catches up instantly.
- *Rejected:* each page keeping its own state, which drifts out of step.

**A tablet controller, the keyboard as backup, a PIN-locked team panel, the laptop's hotspot.**

- *Why:* visitors need buttons, and none can appear on the projection.
- *Rejected:* visitors' own phones, since they would need the network details and university Wi-Fi often blocks devices from reaching each other.
- The PIN keeps visitors out of calibration and weights; it isn't security.

**A plain background by default, clipped to the footprint; an optional offline map.**

- *Why:* the model already has streets, and a drawn map would compete with them, while anything outside the footprint would land on the table.
- The optional map (4 Nov) helps on a screen. It is stored offline so it works without internet, and drawn through the calibration so it lines up.
- *Rejected:* live map tiles, which would need warping to match the projection.

**Street layers on 200 m route pieces, pooled along shared streets, faint when thin.**

- *Why:* colour should follow the model's real streets, not a grid. Pooling fixed one bus standing for a whole street (§7.5), and faint lines show where evidence is thin.
- *Rejected:* removing the street layers. They are the projection's main statement about congestion, and these changes address the flaws found.

**Direction modes, and one line per stretch.**

- *Why:* the owner disliked colours peeking out from under each other, and asked for both ways of showing direction, switchable. One colour (the default) is simplest; side by side shows that inbound and outbound often differ.

**Zones and grid both available.**

- *Why:* which reads better on a physical model can only be judged on the model. Zones match TfNSW's own data; the grid removes the effect of zone size.

**Service gap as a difference of ranks.**

- *Why:* it has no units, and a jobs-to-departures ratio explodes where departures are near zero.

**Pressure from ranks, with neutral fill and jobs excluded.**

- *Why:* ranks resist outliers, neutral fill avoids rewarding missing data, and leaving jobs out avoids counting them twice.
- *Rejected:* min–max scaling, and dropping missing inputs (which pushed empty areas to the top).

**Replay through the same engine, on recorded time.**

- *Why:* one code path means a replay is exactly what live showed, and the 10-minute windows stay correct at any speed.
- *Rejected:* pre-rendered videos, which couldn't change layers during the demo.

**A 25× time-lapse as the showcase default.**

- *Why:* it shows congestion building and clearing within minutes, works at any time of day, and doesn't depend on venue Wi-Fi.

**The replay picker: Replay mode, then a day dropdown, then AM/PM.**

- *Why:* 25 peaks, growing by two each weekday, would fill the tablet as buttons. Opening the picker doesn't change the model, so browsing days is safe.

**The legend reads the engine's thresholds, and doesn't invent ranges for TfNSW's categories.**

- *Why:* a copy of the numbers in the legend would drift from the code. Made-up speed or passenger ranges for TfNSW's categories would mislead.

**Recordings trimmed to the footprint.**

- *Why:* statewide feeds cost about 14 GB in a few hours, and replays only need what can appear on the model.

**A shared daily request budget, full speed only in peaks.**

- *Why:* the quota is about 60,000 requests a day, measured, and constant 15-second polling ran out at 7 pm.
- *Rejected:* fixed 15-second polling.

**Network errors retried within 30 seconds.**

- *Why:* a Wi-Fi drop should cost half a minute, not the five-minute back-off used for "too many requests" errors.

**BOAM joined on stop ID, weekdays only, absolute bands.**

- *Why:* BOAM's trip IDs don't match the timetable. Weekends behave differently. Absolute bands mean days added later are comparable.

**Our congestion measure compares against night-time free flow.**

- *Why:* comparing against the timetable mostly measured timetable padding (§10.1).

**Two sets of thresholds for our measure.**

- *Why:* round, explainable numbers on the model; frequency-matched ones for a fair statistical comparison.

**The timetable refreshed on 30 October.**

- *Why:* the old calendar ended 5 November, the day before the showcase, and every vehicle would have lost its schedule.

**The server kept alive by launchd (3 November).**

- *Why:* the Mac shut down on 3 November, the hand-started server stayed down for seven hours, and a PM peak was lost.

---

## 10. Step 6: our congestion measure compared with TfNSW's

### 10.1 First attempt: delay against the timetable (dropped)

The plan was to measure how much time buses lose against the timetable per 100 m of street. On the 28 October AM peak the median bus *gained* 3–5 seconds per 100 m — even on streets where TfNSW reported severe congestion.

The reason is that bus timetables are padded. Buses are scheduled generously and wait at "timing points" when early, so time against the timetable mostly measured the padding, not the traffic. Variants (stretches of 100, 300 and 500 m, with stops included or excluded) didn't fix it. This is worth a paragraph in the final report as an informative negative result.

### 10.2 Adopted method: travel-time index

The method finally used is the travel-time index against night-time free flow, described in §7.10. It needs no timetable at all.

### 10.3 How agreement was measured

Every recorded weekday peak was replayed through the engine. Each of our readings (a bus's 300 m+ stretch) was paired with the `congestion_level` the same bus reported at the end of that stretch. Agreement was then measured three ways:

- **Per reading:** do the two give the same level for the same bus at the same moment? Measured with exact agreement, agreement within one level, and quadratically weighted κ (§3.3), which counts a one-level miss as mostly agreeing and a three-level miss as not at all.
- **Per segment per 10 minutes:** the same comparison after averaging both within each street piece and 10-minute window.
- **Per street over a whole peak:** do streets that TfNSW calls more congested also have higher TTI on average? Measured with Spearman's ρ (§3.3).

### 10.4 Experiment: which baseline and stretch length agree best?

Run by `scripts/experiment_congestion_collect.py` and `_evaluate.py`. To avoid fitting and scoring on the same data, the thresholds were fitted on Monday–Wednesday peaks and scored on Thursday–Friday.

| Baseline | Stretch | Has a baseline | Fitted thresholds | κ per reading | κ per 10 min | Spearman per street |
|---|---|---|---|---|---|---|
| Night | 300 m | 62% | 2.66 / 2.98 / 3.32 | 0.009 | 0.015 | **0.275** |
| Night | 600 m | 61% | 2.42 / 2.70 / 3.01 | 0.003 | 0.012 | 0.104 |
| Night | 1000 m | 62% | 2.30 / 2.59 / 2.89 | –0.009 | 0.003 | 0.181 |
| Midday | 300 m | 73% | 1.91 / 2.12 / 2.34 | –0.022 | –0.018 | 0.182 |
| Midday | 600 m | 73% | 1.70 / 1.87 / 2.05 | –0.025 | –0.016 | 0.038 |

| Baseline | Stretch | Has a baseline | Fitted thresholds | κ per reading | κ per 10 min | Spearman per street |
|---|---|---|---|---|---|---|
| Midday | 1000 m | 74% | 1.62 / 1.80 / 2.01 | –0.023 | –0.013 | 0.156 |

The night baseline with 300 m stretches did best, so it is the one used.

### 10.5 Final result

Run by `scripts/analyse_congestion.py` over 9 weekday peaks: 190,251 bus stretches, of which 62% had a baseline. The analysis thresholds were used.

| Scale | Result |
|---|---|
| Per reading (97,488 pairs) | Exact agreement 63%, within one level 73%, **weighted κ 0.01** |
| Per segment per 10 min (83,634) | Exact agreement 63%, κ 0.016 |
| Per street, whole peak (1,184 streets) | **Spearman ρ 0.24** |
| Median TTI by TfNSW level | Smooth 1.75 · Stop and go 2.03 · Congested 2.02 · Severe 1.93 |
| Against BOAM time lost (2,207 segments) | Spearman 0.08 |

### 10.6 What it means

**Per bus or per 10 minutes, the two measures don't agree beyond chance.** The 63% "exact agreement" sounds good, but both measures call about 77% of readings "smooth", so they match by coincidence most of the time. κ removes that coincidence and leaves almost nothing.

**Averaged per street over a peak, there is a weak but real relationship** (ρ ≈ 0.24–0.28). Streets TfNSW calls congested do tend to be somewhat slower.

**The median TTI barely changes across TfNSW's levels** (1.75 for smooth, about 2 for the others). Buses TfNSW labels "severe" aren't, on average, moving slower than those labelled "stop and go".

Likely reasons:

1. The night baseline has fewer passengers boarding, so every peak street reads somewhat slower.
2. A single bus's 300 m reading is noisy: one long stop changes it a lot.
3. TfNSW's label is undocumented and per vehicle. It may not be a measure of the street that bus just drove at all.
4. BOAM time lost is measured against the timetable (so it is affected by padding), and three of its days are school holidays.

**Conclusion for the final report:** the two capture different things and only line up when averaged per street over a peak. The model shows both, labelled as such, and the tablet legend explains what each one means.

This analysis ran before street pooling (4 November). Pooling changes what the model draws, not these per-reading statistics. Rerunning with the seven weekday peaks recorded since would firm up the per-street figure (§16).

Outputs are in `data/analysis/`:

- `congestion_agreement.json`;
- `congestion_disagreements.geojson` (337 segments where the two differ by 2 or more levels);
- `experiment_results.json`;
- `freeflow.json`.

## 11. Testing, validation and performance

### 11.1 Automated tests

There are 208 automated tests (`pytest`), which run in about 2 seconds without internet access. They cover:

- the timetable filter and map matching;
- punctuality, interchange counts and statistics;
- calibration maths and footprint trimming;
- the display-state rules (every allowed and rejected value, including the corner-fit offsets and their saving across restarts);
- street pooling: same street, separate directions, parallel and crossing streets, order independence, both-direction values, the towards-city flag;
- replay: pacing, looping, coverage, jumping, pausing, choosing between two recorders' files;
- area maths, including the per-fifth ranges, and the pressure score;
- BOAM maths, including both-direction pooling;
- the request budget, the travel-time tracker and the agreement statistics;
- the teammate's stop tracker, history, trip-update and route-focus code;
- the background map build: layers, labels, tunnels, multi-part water outlines.

**What the tests don't cover:** the projector and tablet pages' JavaScript. Those are checked by eye in Chrome after every change.

### 11.2 Live checks

- TfNSW field coverage on 23 October: §5.1.
- After the 30 October timetable refresh, all 324 live vehicles matched a route name.

### 11.3 Visual checks in Chrome

Checked by eye:

- the projector on every layer;
- all four street-direction modes, and BOAM, on a paused 4 November AM replay;
- the background map in Plain, Detailed and Full, in both themes;
- the tablet's three tabs;
- the replay picker: a full day, a 66% day, days missing a peak, and the open and closed states;
- every legend combination: each lens, live and BOAM streets, each area layer on zones and grid, stop history.

Problems found and fixed this way:

- lines spilling past the model's edge;
- a hidden row showing;
- the recordings list missing after startup;
- the nudge buttons' layout;
- the harbour vanishing from the map;
- a clipped tablet button;
- "TfNSW" lower-cased in a heading;
- a raw date (20261110) in the service-gap note;

- "Live now" staying lit while Replay mode was open.

### 11.4 Performance

| Measure | Before | After |
|---|---|---|
| Live server processor use | ~85% | 1–18%, averaging ~7%, at the 5 Nov PM peak with 487 vehicles |
| Replay throughput, dense data | ~11× real time | ~96× |
| 50× replay, measured | — | 50.0× |
| Static layers: build / from cache | — | ~5 s / 0.02 s |
| Street snapshot with pooling | 1.4 s (first version) | ~40 ms (~110 ms once at startup) |
| BOAM import size | ~400 MB per day | ~2.7 MB per day |
| Daily requests | Quota ran out at 7 pm | ~57,000 a day, peaks at full rate |
| `/stop_history` full recompute | ~13 s at 586k visits (3 Nov) | ~2 s at 866k visits (5 Nov); 0.03 s cached |

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# Part IV — Running it

## 12. Operating it

### 12.1 Starting and stopping

The server runs as a launchd job (`com.pulsecbd.server`) installed on 3 November. It starts when the owner logs in, and restarts within 30 seconds whenever it stops. It runs under `caffeinate`, which stops the Mac idling to sleep while it runs.

```
python scripts/install_server_service.py                 # install the job and start the server
launchctl kickstart -k gui/$(id -u)/com.pulsecbd.server  # restart, e.g. after a code change (~30 s down)
python scripts/install_server_service.py --uninstall     # stop for good (a plain `kill` only restarts it)
```

- Because it is a per-user job, after a reboot someone must log in before the server starts.
- Startup takes about 15 seconds. The area and BOAM layers finish building in the background a few seconds later, and their buttons are greyed out until then.
- **Projector:** open `http://127.0.0.1:8765/` full-screen on the laptop.
- **Tablet:** open `http://<laptop address>:8765/control`; the address is printed in the log.
- After any code change, reload both pages.
- For a one-off run in a terminal instead, uninstall the job first (otherwise the port is taken), then run `python scripts/run_publish.py`.

### 12.2 Recording

**The server records everything it downloads while live** (files tagged `serve`).

**A second launchd job, `com.pulsecbd.recorder`, records the weekday peaks even when the server isn't running** (files tagged `rec`).

- It runs at 06:55 and 16:25 for 130 minutes each, so it covers each peak with five minutes either side.
- When the server is already polling, the recorder sees the server's "heartbeat" file (`data/recordings/.server_polling`) and stands aside, so the same peak isn't paid for twice. Its log then says "0 feed payloads received", which is normal.
- It takes over if the server stops.

```
python scripts/run_record.py --minutes 130          # one-off recording
python scripts/install_recording_schedule.py        # weekday 06:55 and 16:25 (installed)
python scripts/install_recording_schedule.py --uninstall
```

**Where files go.** Recordings are stored as `data/recordings/{mode}/{feed type}/{hour}.{serve|rec}.ndjson.gz`. Each line is one download, with its fetch time.

- Hours are in UTC, not Sydney time: `20261104T15` is 2 am on 5 November in Sydney.
- If both writers recorded the same hour, replay reads the larger file. Until 30 October it preferred the `rec` file, which hid most of the Thursday 29 October PM peak.

**Sleep.** The wake schedule (`scripts/wake_schedule.sh`) was never needed, because sleep is disabled.

### 12.3 Staying within the data quota (`engine/sources/budget.py`)

The server and the recorder each add every request they make to one shared count for the day (`data/api_usage.json`, locked so the two processes can't both write at once). The budget is set to 57,000, about 5% under the measured quota. How fast the server polls then depends on the time of day:

- **During peaks** (weekdays 06:55–09:05 and 16:25–18:35), it polls every 15 seconds: 15 requests per cycle, one per feed.
- **Between peaks**, it holds back the requests the rest of today's peaks will need, and spreads what's left evenly until midnight. The interval stretches from 15 seconds up to at most 600 seconds.
- **When the day's budget is spent**, it polls once every 600 seconds: a few hundred requests a day. The live view slows down instead of going dark.

Current usage and pace are under Team → System health on the tablet, and at `/health`.

### 12.4 Calibration: lining the image up with the model

**Why it's needed.** Without calibration, the projector page simply fits the footprint rectangle into the screen. On the real model, the projector's position and angle mean that rectangle won't land exactly on the model's streets.

**First-time calibration:**

1. Press **C** on the laptop, or **Start** under Team → Alignment on the tablet. Four or more draggable crosshairs appear on the projection.
2. Drag each crosshair onto a physical mark on the model whose latitude and longitude are known (a corner, a distinctive intersection).
3. Type in that mark's latitude and longitude.
4. Save. The server computes a homography (§3.3) and saves it to `data/calibration.json`. The projection switches to it immediately, and it survives restarts.
5. Press **V** for the verification overlay. Route lines are drawn bright; check they sit on the model's streets and rail lines before trusting the projection.

**Small drift** between weekly sessions is fixed with the **nudge** on the team panel: shift by 2 px or 20 px, and scale.

**Corner fit** (teammate's, merged 5 November) is the quick way to line the map up when the projector has moved or isn't square to the table: Team → Alignment → Corner fit, or **F**.

- Four handles appear on the footprint's corners (north-west, north-east, south-east, south-west).
- Drag each onto the matching corner of the model, and the whole map follows. It is a homography through those four points, so it also removes keystone (the trapezoid shape from a projector that isn't straight above).
- For fine moves, set "Arrows move" to a corner and use the arrow pad.
- **Reset corners** puts them back.

The nudge and the corner offsets are both saved in `data/projection_nudge.json` and survive restarts. Saving a new crosshair calibration resets both.

According to the teammate's README update of 5 November, the projection is now aligned on the physical model. The crosshair calibration itself is recorded as tested only with synthetic points (§15).

### 12.5 Weekly model session

1. Check the server is running; open both pages.
2. Team → verification overlay on. Do the corridors sit on the model's streets?
3. Small drift: nudge. The projector moved or isn't square: corner fit. Neither enough: recalibrate with at least four marks.
4. Try every layer on the model. Note:
    - whether colours read on the white model;
    - whether line widths are visible (the glow that hid small drift is gone);
    - zones versus grid;
    - whether Full map labels are readable.

### 12.6 Showcase-day checklist

1. **Laptop:** on **power**, hotspot on, tablet joined to it, `/control` open and **reloaded**, team PIN entered once.
2. **Projector page:** full-screen and reloaded. Status line off, or on to label replays.
3. **Alignment:** verification overlay check, nudge if needed, then turn the overlay off.
4. **Defaults to confirm:** background map Plain, street directions One colour, no layers on.
5. **Replay for the presentation:** Replay mode → day → AM or PM peak, at 25×. Good candidates are any weekday peak at 100%, such as Wednesday 4 November AM, Thursday 5 November AM, or Monday 26 or Wednesday 28 October AM or PM.
6. **Live:** use it only if Team → System health shows fresh feeds (a few seconds old) and the API budget isn't spent.
7. **After each demonstration, press Live now.** Polling and recording are paused during replays.

### 12.7 Troubleshooting

| Symptom | Likely cause | What to do |
|---|---|---|
| Tablet header says "connecting…"; projector frozen | Server not running or restarting | `lsof -nP -iTCP:8765 -sTCP:LISTEN`; if nothing, `launchctl kickstart -k gui/$(id -u)/com.pulsecbd.server`; read the end of `data/recordings/server.log` |

| Symptom | Likely cause | What to do |
|---|---|---|
| Tablet can't open the page at all | Not on the same network | Join the laptop's hotspot; use the address printed in the server log |
| A page looks like an older version | It has old code cached | Reload the page |
| Area or BOAM buttons greyed out | Layers still building after startup, or data files missing | Wait a few seconds after a restart; check the log for "areas" or "boam" errors |
| Background map buttons greyed out | `data/basemap/basemap.json` missing | `python scripts/fetch_basemap.py` (or `--cached`), then restart |
| Vehicles stop moving while live | Wi-Fi dropped, or polling has slowed to save the quota | Team → System health: feed ages of minutes mean no data is arriving; a long polling interval means the server is pacing itself (outside peaks) or the day's budget is spent. Switch to a replay |
| Street layer nearly empty | Only vehicles from the last 10 minutes count; just after startup or switching to Live there are few | Wait a few minutes, or use a replay (jumping fills the 10 minutes before the target) |
| A replay looks sparse | It is a partly recorded session | Check the coverage note under Recorded peak; choose a peak at 98–100% |
| The model shows a replay after the demo | Replays don't end by themselves; they loop | Press Live now |
| Colours are offset from the streets | Calibration drift | Nudge (Team → Alignment), or recalibrate |
| Server keeps restarting | It crashes at startup | Read `server.log`; run `python scripts/run_publish.py` in a terminal after uninstalling the job, to see the error |

### 12.8 Maintenance commands

```
# refresh the timetable (then restart the server)
python scripts/run_phase1.py --force-refresh
# import new BOAM day files, keeping only footprint rows
python scripts/import_boam.py --delete-originals ~/Downloads/BOAM_*.txt
# export BOAM sections to data/boam/sections.csv
python scripts/export_boam.py
# rebuild the night free-flow baseline (~5 min), then restart
python scripts/build_freeflow.py
# step 6 analysis (~15 min)
python scripts/analyse_congestion.py
# only if zones or postcodes change
python scripts/prepare_demand_data.py --poa-dir <ABS shapefile dir>
# background map from OpenStreetMap (--cached: rebuild from the saved download), then restart
python scripts/fetch_basemap.py [--cached]
```

---

## 13. The code repository

The code is in a private GitHub repository shared by the team, `0rangeGoose/Civil3704-group-6`, known locally as the remote `group-repo`.

| Branch | Where | What it holds |
|---|---|---|
| `w12merge` | local + GitHub | **Current, and the build to run.** The owner's work merged with the teammate's (merge `3ab2a8a`, 30 Oct), plus everything since (see below) |
| `yosephW12` | local + GitHub | The owner's work through 30 October, before the merge (`3bfd432`) |
| `integration` | GitHub, `67d4d9c` | The teammate's 11 commits of 29–30 October on top of the owner's work to 26 October. Fully contained in `w12merge` |
| `test` | local + GitHub | The owner's prototype as of 28 August |
| `main`, `mid-project-demo`, `test-pipline`, `Mid-project-demo-A-edits` | GitHub | The teammates' template and dashboard work |

Commits on `w12merge` since the merge:

| Commit | What it did |
|---|---|
| `6ee94d2` | Stop visits recorded only while live |
| `17ad8fc` | The server's launchd job |
| `6fcca03` | Background map (Plain, Detailed, Full) |
| `14b7b9c` | Street pooling, faint streets, direction modes |
| `95c722a` | Replay day picker and colour definitions |
| `b456295`, `2494941`, `e6d99d9` | Teammate (4–5 Nov): corner fit, merged into `w12merge`, and a README rewrite; also adds `pulse-cbd-context.md`, their handoff notes |
| (this commit) | Merge of the above with this report and the owner's README update; the teammate's README kept, as it already included those updates |

**Which branch is the showcase build** still has to be agreed with the teammate (§18). `w12merge` is the only one with both sides' work and everything since.

**Clean-up before submission.** The teammate committed:

- personal editor settings (`.claude/settings.json`, `.vscode/settings.json`);
- about 55 MB of PDFs and a spreadsheet in `template-repository-main/New folder/`, not stored with Git LFS;
- their individual report, at the repository root.

**Security.** A teammate's TfNSW API key is committed in `template-repository-main/src/file.env` (commit `d85b5e1`). The repository is private, but the key should still be regenerated. The owner's key is only in `.env`, which Git ignores; it appears in no commit, and this is checked before every push.

---

## 14. State of the laptop (5 November, 4:30 pm)

| Item | State |
|---|---|
| Server | launchd job, live, 487 vehicles, feeds 8–10 s old, 2 pages connected. It runs `95c722a`: **corner fit needs a server restart and page reloads** to appear (deliberately not done during the PM peak recording) |
| Recorder | Installed; the 16:25 run started on time and is standing aside while the server polls |

| Item | State |
|---|---|
| Sleep | Disabled (`sudo pmset -a disablesleep 1`). Undo after 6 November with `sudo pmset -a disablesleep 0`. Keep the Mac **on power**: it was found on battery on 3 November |
| API usage | About 57,000 on every full day from 29 October to 2 November, and on 4 November. 3 November: 39,663 (server down 11:30–18:33). 5 November: 33,450 by 4:27 pm, polling every 15 s (peak) |
| Disk | 68 GB free. Recordings 27 GB; stop visits 128 MB; analysis 58 MB; BOAM 42 MB; background map 16 MB |
| Server log | `data/recordings/server.log`: 131 MB, about a million lines, never trimmed |
| Node.js | Installed via Homebrew (only for syntax checks) |

**Replayable sessions** (share of the expected bus updates actually recorded):

| Day | AM | PM |
|---|---|---|
| Thu 5 Nov | 100% | recording now |
| Wed 4 Nov | 100% | 99% |
| Tue 3 Nov | 100% | lost (Mac shut down) |
| Mon 2 Nov | 100% | 98% |
| Sun 1 Nov | 66% | 66% |
| Sat 31 Oct | 66% | 66% |
| Fri 30 Oct | 99% | 100% |
| Thu 29 Oct | 100% | 99% |
| Wed 28 Oct | 100% | 100% |
| Tue 27 Oct | 100% | 86% (Wi-Fi drop ~5 pm) |
| Mon 26 Oct | 100% | 100% |
| Sun 25 Oct | 84% | 96% |
| Sat 24 Oct | 100% | — |
| Sun 6 Sep | 54% | 99% |
| Sat 22 Aug | — | 44% |

Weekends read 66% because polling slows outside weekday peaks to save the quota. "Latest 2 hours" is also always available.

**Known outages:**

- Wi-Fi drops: 23 October from about 4:31 pm (most of that PM peak lost), the afternoon of 25 October, and 27 October from 5:00 to 5:20 pm and from 7:40 to 10:42 pm.
- TfNSW throttled the key from 2 to 4 pm on 28 October (not our quota).
- The Mac shut down after 11:30 am on 3 November, and the server stayed down until 6:33 pm.

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# Part V — Assessment

## 15. Limitations and caveats

These should be stated plainly in the final report.

**Data**

- **TfNSW's fields are undocumented categories.** Neither the traffic nor the crowding field has published numbers behind it. Crowding is almost always "seats available" off-peak. Light rail and ferries report neither field. Sydney Trains' traffic is always unknown, and about 70% of Sydney Trains have no usable delay.
- **BOAM's limits.** Its "actual" times have been TfNSW's live predictions since March 2026. People who don't tap aren't counted. Load assumes the occupancy given at a stop is the load departing it. Three of its 13 weekdays are school holidays.
- **Opal 2016 is old.** It is ten years old and predates the Metro and the CBD light rail. Bus taps are placed only by postcode.
- **Recordings have gaps** wherever Wi-Fi dropped (§14).

**Method**

- **Relative layers.** The service gap and pressure are relative to the footprint, not measured against planning standards. Jobs and Opal fifths are affected by zone size: a large zone can rank high just for being large. The grid avoids this.
- **Departures measure reachable service.** A trip counts once per stop within reach, not once in total.
- **Thin street data.** At the AM peak, 11% of CBD streets and 32% beyond 4 km still rest on one bus (drawn faint); off-peak far more do.
- **Pooling is by geometry.** Streets are matched by shape, not from a real street network.
- **One colour hides direction.** The default view averages both directions, hiding differences between inbound and outbound traffic; Side by side or To/From city shows them. "Towards the city" means towards Town Hall, which is close to arbitrary inside the CBD.
- **Simplified punctuality.** The thresholds are simplified defaults (§7.2), and early running counts as on time.
- **Our congestion measure is limited.** It uses a night baseline (fewer passengers, so every peak street reads somewhat slower). 38% of peak readings have no baseline. Its agreement with TfNSW is weak (§10).

**Setup**

- **Limited testing on the physical model.** The teammate reports the projection aligned on the model with corner fit; checks of every layer there aren't recorded. Crosshair calibration is recorded as tested only with synthetic points. The glow that used to hide small drift is gone.
- **Replays aren't labelled.** The projection doesn't say it is showing a replay unless the status line is on.
- **Weak access control.** The team PIN is checked only in the browser, and the server accepts connections from any device on the network (use the hotspot).
- **One laptop.** Replays depend on one laptop and one disk, and the server log is never trimmed.
- **Untested page code.** The projector and tablet pages have no automated tests.

---

## 16. Remaining work, in priority order

**Before the showcase (today and tomorrow morning)**

1. **Restart the server and reload both pages** after today's PM peak (18:35), so the running build includes corner fit. Then **finish checking on the physical model** (alignment is reported done by the teammate):
    - every layer;
    - zones versus grid;

    - colours on the white model;
    - line widths without the glow;
    - Full map label size.

2. **Test the networking.** A Mac can't run a hotspot and join Wi-Fi at the same time, so live mode needs one of:

    - Ethernet;
    - USB tethering to a phone;
    - a phone hotspot that both the laptop and the tablet join.

    If none works, run replays only; they need no internet.

3. **Agree with the teammate** that `w12merge` is the showcase build, and that about 70% of Sydney Trains show grey.

4. **Keep recording** today's PM peak: Mac plugged in and on Wi-Fi until 18:35.
5. **Back up the best 3–4 peaks** to an external drive. Replays depend on this one disk.
6. **Pick the showcase replay** (§12.6), and rehearse the tablet flow: Replay mode → day → peak → Live now.
7. **Make it clear when a replay is showing:** status line on, or say so out loud.
8. **Make the report figures:**
    - service gap, pressure, BOAM load and time lost;
    - the step 6 tables and the map of disagreements;
    - the colour table in §8.
9. Ask the teammate to **regenerate the committed API key.**

**Optional or undecided**

- Rebuild the free-flow baseline and rerun the step 6 analysis with the peaks since 30 October (`build_freeflow.py` → restart → `analyse_congestion.py`, about 20 minutes).
- Leave the three holiday BOAM days out of the averages (they barely change results: loads match to about one percentage point).
- Add an alert on the Mac when Wi-Fi stalls (offered, not built).
- Add an "early" punctuality state; cite TfNSW's official on-time definitions.
- Move "Traffic (our measure)" to the team panel. It disagrees with TfNSW's layer, and at peak it reads "slow" almost everywhere.
- Make the route-filter lines match the no-glow street style (offered).
- Trim or rotate `server.log`.

**After the showcase:** `sudo pmset -a disablesleep 0`; `python scripts/install_server_service.py --uninstall`; `python scripts/install_recording_schedule.py --uninstall`.

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## 17. Suggestions for improvement

1. **Guided demo mode:** one button plays a scripted tour (AM peak at 25× → traffic → service gap → pressure), with captions on the tablet.
2. **A simpler visitor tab:** move BOAM, the grid switch, street directions and our own measure behind a "More" button. The legend gets long when several layers are on.
3. **A replay label on the projection itself:** the date and time in a corner, independent of the status line.

4. **Better step 6 analysis:**
    - a baseline by time of day, from weekday shoulder hours;
    - comparison per route rather than per piece;
    - excluding the time spent at stops;
    - a repeat of the analysis on pooled streets.
5. **Check the service gap against an accessibility measure,** such as jobs reachable within 30 minutes, for sample zones. Also test how sensitive it is: walking distance of 400 m instead of 200 m, and each pressure weight set to 0.
6. **Use a real street network** (OpenStreetMap is already downloaded) instead of pooling by geometry.
7. **Better BOAM measures:** passenger-kilometres rather than mean load, and term-time days only.
8. **Faster startup,** by caching route corridors like the area layers.
9. **Browser tests** for the tablet and projector pages, now that the legend and replay picker hold real logic.
10. **Touch markers on the model** at stations (from the original brief; out of scope this term).

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## 18. Open decisions waiting on the owner

1. Confirm `w12merge` as the showcase build, with the teammate.
2. Which peak is the showcase's default replay?
3. Leave the holiday BOAM days out of the averages?
4. Build the Wi-Fi stall alert?
5. Move "Traffic (our measure)" off the visitor tab?
6. Rerun step 6 with the newer peaks before writing the final report?

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## 19. Timeline

| Date | What happened |
|---|---|
| August | Prototype built in six phases on `test`: data engine, projector page, calibration, statistics |
| 22–23 Aug | Proposal and timeline documents |
| 25 Sep | Design session; plan, decisions and build order agreed (§2.3) |
| 23 Oct | Steps 1–3: footprint, tablet controller, recording, street traffic and crowding, replay. The rate-limiting problem and slow map matching fixed. The first scheduled PM peak lost to Wi-Fi |
| 23 Oct | Steps 4–5: area layers and pressure. BOAM imported (5 days) and used three ways. Branch `integration` pushed |
| 24–25 Oct | Server run continuously. Quota exhausted at 7 pm on Sunday, which is how the quota was measured |
| 26 Oct | Quota-aware polling and 30-second network retries. The first complete weekday peaks |
| 29–30 Oct | The teammate pushes 11 commits to `integration` |
| 30 Oct | Replay pause and seek. Timetable refreshed. Step 6: new method, experiment, final analysis. 12 more BOAM days (school term). Work pushed to `yosephW12`. Replay file-choice fix (Thursday 29 October PM from 50% to 99%). Merged with `integration` as `w12merge`. Stop visits made live-only |
| 3 Nov | Mac shut down after 11:30. Server down until 18:33 and the PM peak lost. Server moved to a self-restarting launchd job |
| 4 Nov | Optional background map (Plain, Detailed, Full). Street glow removed. Streets pooled, faint when thin; long BOAM sections left out of time lost. Street direction modes, with one line per stretch |

| Date | What happened |
|---|---|
| 4–5 Nov | Teammate: corner fit for aligning the projection on the model (F), merged into `w12merge`; README rewrite |
| 5 Nov | Tablet: Replay mode with a day dropdown and AM/PM buttons; every colour defined with its cut-off, read from the server; stop-history legend. This report rewritten for readers without prior context, and the teammate's work merged in. Interchange pulses removed from the projection |

## 20. Key files

| Area | Files |
|---|---|
| Configuration | `config/config.yaml` (Appendix C) |
| Fetching and recording | `engine/sources/live.py` (polling), `budget.py` (quota), `recorder.py`, `footprint.py` (trimming), `replay.py` (sessions and playback) |
| Processing | `engine/normalise/build.py`, `enrich.py` (joining to the timetable), `map_match.py`, `adherence.py`, `trip_updates.py`, `levels.py`, `segments.py` (streets and pooling), `travel_time.py` (our measure), `aggregate.py` (statistics, interchanges) |
| Static layers | `engine/demand/areas.py` (jobs, departures, service gap, Opal), `boam.py`, `pressure.py` |
| Server | `engine/publish/server.py` (addresses), `feed.py` (live state), `control.py` (display state), `calibration.py`, `homography.py`, `route_focus.py`, `shapes_export.py` |
| Stop history (teammate) | `engine/history/stop_tracker.py`, `store.py` |
| Pages | `renderer/index.html` (projector), `renderer/controller.html` (tablet) |
| Scripts | `run_publish.py` (server), `run_record.py`, `install_server_service.py`, `install_recording_schedule.py`, `fetch_basemap.py`, `run_phase1.py` (timetable), `prepare_demand_data.py`, `import_boam.py`, `export_boam.py`, `build_freeflow.py`, `analyse_congestion.py`, `experiment_congestion_collect.py`, `experiment_congestion_evaluate.py` |
| Tests | `tests/` (208 tests; e.g. `test_segments.py`, `test_boam.py`, `test_areas.py`, `test_control.py`, `test_replay.py`, `test_basemap.py`) |
| Data (not in Git) | `data/static_cache/` (timetable), `data/recordings/`, `data/demand/`, `data/opal/`, `data/boam/`, `data/analysis/`, `data/history/` (stop visits), `data/basemap/`, `data/api_usage.json`, `data/projection_nudge.json`, `data/calibration.json` |
| Documents | `README.md`, `docs/DECISIONS.md`, `docs/PROJECT_REPORT.md` |

## Appendix A. Display settings (`engine/publish/control.py`)

Every setting returns to its default when the server restarts, except the nudge and the corner-fit offsets, which are saved to `data/projection_nudge.json`.

| Setting | What it controls | Allowed values | Default |
|---|---|---|---|
| `theme` | Projection colours | dark, light | dark |
| `basemap` | Background map | plain, detailed, full | plain |
| `street_directions` | How street layers show directions | combined, split, inbound, outbound | combined |
| `verify` | Verification overlay | on / off | off |
| `hud` | Status line on the projection | on / off | off |

| Setting | What it controls | Allowed values | Default |
|---|---|---|---|
| `calibrate` | Calibration crosshairs | on / off | off |
| `stops`, `stop_history` | Stop markers; stop history colours | on / off | off |
| `stop_metric` | Stop history delay statistic | median, mean | median |
| `routes` | Routes singled out by the filter | up to 8 route names | none |
| `route_slots` | Colour assigned to each filtered route | 0–7, set by the server | — |
| `lens` | What vehicle colour shows | adherence, congestion, crowding | adherence |
| `line_layer` | Street layer | none, congestion, derived, crowding, boam_load, boam_lost | none |
| `area_layer` | Area layer | none, jobs, service_gap, opal, pressure | none |
| `area_unit` | Area shape | zones, grid | zones |
| `boam_period` | Peak used by BOAM layers and pressure | am, pm | am |
| `weights` | Pressure inputs: congestion, crowding, service_gap, opal, boam_load | 0–3 each | 1 each |
| `playback` | Live or a session (e.g. `2026-11-04-am`), speed, paused | live or a recorded session; 1–100×; yes/no | live, 25×, not paused |
| `modes` | Vehicle groups: trains, metro, lightrail, buses, ferries | on / off each | all on |
| `nudge` | Alignment correction | dx and dy ±2000 px; scale 0.5–2 | 0, 0, 1 |
| `corners` | Corner-fit handles shown | on / off | off |
| `warp` | Corner-fit offset of each footprint corner (tl, tr, br, bl) | x and y ±2000 px each | all 0 |

## Appendix B. Server addresses (`engine/publish/server.py`, port 8765)

| Address | What it returns |
|---|---|
| `/` | The projector page |
| `/control` | The tablet page (`/stats` redirects to its Stats tab) |
| `/ws` | The WebSocket: vehicle positions about twice a second and the display state; accepts setting changes |
| `/config` | Footprint, polling timing, mode groups, level labels, punctuality and TTI thresholds, whether a background map exists, the team PIN |
| `/health` | Vehicles per mode, feed ages, API use today, polling interval, playback |
| `/snapshot` | Every current vehicle |
| `/shapes` | Route lines for drawing |
| `/segments` | Street pieces with pooled values, counts, both-direction values and the towards-city flag |
| `/boam` | BOAM sections (`?meta=1`: the days used) |
| `/areas` | Zones and grid squares with values and colour bins (`?meta=1`: availability, build details, per-fifth ranges) |
| `/pressure` | The live combined score per area for the current shape and weights |
| `/stops`, `/stop_history` | Stop markers; per-stop delay and crowding summaries |

| Address | What it returns |
|---|---|
| `/routes`, `/route` | The route list for the filter; one route's line and stops |
| `/recordings` | Replayable sessions with coverage |
| `/basemap` | The offline map layers and labels |
| `/aggregate` | The Stats tab's figures |
| `/calibration` | Read the calibration; send at least 4 points to save a new one |

## Appendix C. Configuration (`config/config.yaml`)

| Section | What it sets |
|---|---|
| `bounding_box` | The footprint corners (§2.2) |
| `calibration_path` | Where the calibration is saved |
| `static_gtfs` | Timetable download address and cache location |
| `realtime_gtfs` | The 15 live feeds polled (7 vehicle-position, 7 trip-update, 1 combined alerts), the polling interval, and `stale_after_seconds` (90 s: how long without an update before a vehicle is treated as gone) |
| `adherence_thresholds_seconds` | Per-mode on-time thresholds and the "very late" multiplier (§7.2) |
| `interchanges`, `interchange_radius_m` | The nine interchanges and the 300 m radius counted for the Stats tab (§7.3) |
| `server` | Listens on every network interface, port 8765 |
| `controller` | The team PIN (`3704`) |
| `api_budget` | Daily limit 57,000, the peak windows, the 600 s slowest interval, file locations (§12.3) |
| `derived_congestion` | Free-flow baseline file, model thresholds (1.5 / 2 / 3), analysis thresholds (2.66 / 2.98 / 3.32) (§7.10) |